

Experiment Pipeline — Six Verification Steps

Synthetic dataset, ordering benchmark, ClassBench-ng, and compiled kernel

Generate

10,000 LRF policies · Small / Medium / Large · seed 2025

Convert

LRF → TRF · Algorithm 1 · Ordering 4 (default)

Verify Step 1 — Semantic Fidelity

10,000 policies · 73.12 M evaluations · 0 discrepancies

Verify Step 2 — Anomaly Detection

7,000 trials · 14 configurations · Z3 oracle on 10% sample

Verify Step 3 — Scalability & Depth

$n = 5 \rightarrow 400$ · depth $d = 4$ · two platforms

Verify Step 4 — Ordering Comparison

12 orderings · 10 independently generated policies per size · $n \in \{50, 100, 200, 400\}$

Verify Step 5 — ClassBench-ng

8 rulesets · 32 cases · vs FDD-style fixed-order ablation / HiCuts

Verify Step 6 — Cython Kernel

5 trials · compiled native C · same algorithm, different runtime

Verified Results

- ✓ **Fidelity = 100%**
73,120,887 evaluations, FN = 0 (95% UCI $\leq 4.10 \times 10^{-8}$)
- ✓ **Recall = Precision = 100%**
20,868 TP, 0 FP, 0 FN (pairwise anomalies, Definition 5)
- ✓ **$d = 4$ constant**
all $n = 5 \rightarrow 400$ and all 32 ClassBench-ng cases (by construction; no early collapse)
- ✓ **Cython 39x-41x speedup**
67-177 ns/packet ($n = 25 \rightarrow 400$); runtime, not algorithmic, comparison
- ✓ **Theorems 1-3, Propositions 1-4**
consistent with every measurement (proofs are deductive; experiments are checks)